Predicting Trust and Distrust in Bitcoin Web-of-Trust Networks

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Group 3

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Abstract

Predicting the sign of edges in signed directed graphs is a fundamental challenge in network analysis, with direct applications to fraud detection and risk assessment in decentralized financial systems. We study sign prediction on two Bitcoin web-of-trust networks, Bitcoin OTC and Bitcoin Alpha, framing the problem as binary edge-sign classification under severe class imbalance. We construct a fourteen-dimensional feature vector grounded in Balance Theory, Status Theory, and reciprocity, computed via a streaming pipeline that strictly enforces temporal integrity, and train a Random Forest classifier on an 80/20 temporal split. An ablation study reveals that reciprocity is the most impactful feature pillar (-0.032 AUC on Alpha when removed), followed by Balance Theory, while Status Theory contributes marginally. The full model achieves AUC-ROC of 0.8665 on Alpha and 0.9106 on OTC. Despite the severe class imbalance, the classifier successfully identifies the majority of distrust edges, achieving $F1^-$ of 0.49 on Alpha and 0.64 on OTC, metrics that reflect a genuine detection capability rather than a trivial majority-class bias, as a model predicting trust for all edges would yield $F1^- = 0$. A Signed Graph Convolutional Network (SGCN) trained on the same splits reaches AUC of only 0.5478 and 0.6578 respectively, substantially below the feature-based approach, a gap we attribute to graph sparsity and severe class imbalance in the negative aggregation channel. Cross-dataset transfer yields AUC above 0.87 in both directions, providing a positive answer to our second research question: trust dynamics learned on one platform generalize to the other without retraining.

1 Introduction

Decentralised peer-to-peer platforms such as Bitcoin rely on web-of-trust mechanisms in which users assign explicit positive or negative ratings to counterparties. These ratings form a *signed directed graph* whose negative edges carry critical information about potential fraud, adversarial behaviour, and systemic risk. Despite this, the majority of existing graph-based models either discard edge polarity entirely or treat positive and negative links as structurally equivalent, discarding a rich source of signal that is particularly consequential in settings where identity cannot be verified.

We focus specifically on *sign prediction*: given that an interaction occurs between users u and v, can we predict whether the resulting edge will represent trust $(+)$ or distrust $(-)$? Accurate solutions could underpin early-warning systems for harmful actors in decentralised financial networks. We frame the problem as binary edge-sign classification on two Bitcoin web-of-trust networks that exhibit severe class imbalance. Our approach combines a feature-engineering pipeline grounded in Balance Theory, Status Theory, and reciprocity with a Random Forest classifier trained on temporally split data to prevent leakage. We further compare this against a Signed Graph Convolutional Network (SGCN), which learns node representations through separate aggregation channels for positive and negative neighbourhoods. An ablation study quantifies the marginal contribution of each theoretical feature group, and a cross-dataset evaluation tests whether learned trust dynamics generalise across platforms. Models are assessed via AUC-ROC and negative-class F1-score, metrics chosen for their robustness to class imbalance.

Research Questions. The entire investigation is structured around two research questions:

RQ1. To what extent do structural features derived from Balance Theory and Status Theory predict the sign of new edges in Bitcoin trust networks, and how do these hand-crafted features compare against representations learned automatically by a SGCN?

RQ2. Do trust dynamics learned on one platform (Bitcoin OTC) generalise to an independent network (Bitcoin Alpha), or does the model overfit to platform-specific patterns?

2 Related work

Sign prediction in signed networks has attracted growing interest as a natural extension of classical link prediction. The theoretical foundations of the field trace back to Cartwright and Harary [8], who formalised Balance Theory in social psychology, establishing that triadic relationships tend toward sign-consistent configurations where the product of edge signs is positive. These principles were later operationalised for computational analysis of signed networks by Leskovec et al. [1], who introduced Balance Theory and Status Theory as complementary frameworks to explain the formation of positive and negative edges in online communities such as Epinions and Slashdot, demonstrating that structural patterns derived from triads are strong predictors of edge sign. Their empirical findings constitute the theoretical grounding for our sociological feature pillars and directly motivate our use of signed networks from a similar domain.

The triad classification scheme used in our analysis follows Davis and Leinhardt [7], who systematically categorised directed triangles in small groups, providing the combinatorial vocabulary we use to enumerate balanced and unbalanced configurations in Section 3.2.3.

Feature-based machine learning approaches have since formalised these social-theoretic intuitions. Wu et al. [2] applied a Random Forest classifier to edge sign prediction on the Wikipedia signed network, assembling hand-crafted features from degree statistics and triad counts. Their results showed consistent improvements over logistic regression baselines, validating the utility of sociological features in a supervised classification setting. Although their domain differs from cryptocurrency networks, the core problem structure of binary classification under severe class imbalance transfers directly to our setting. Their work validates the use of a Random Forest in our pipeline, implemented via scikit-learn [9], which provides the `GridSearchCV` and `StratifiedKFold` utilities used in our hyperparameter search.

Graph neural network methods have more recently superseded purely feature-based approaches. Derr et al. [3] proposed the SGCN, which extends spectral convolution to signed graphs by maintaining separate aggregation channels for positive and negative neighbourhoods. This architecture encodes balance-theoretic assumptions directly into the learned representations, and has become the standard deep learning baseline for sign prediction tasks. Our implementation relies on PyTorch Geometric [4], a graph deep learning library that provides an efficient SGCN module, on the SNAP repository [5] from which both datasets are drawn, and on NetworkX [6] for graph construction and feature computation. While existing work evaluates models across multiple signed networks, each model is typically trained and tested within the same platform under random splits. Cross-dataset transfer, that is, training on one network and evaluating on another without any retraining, remains largely unexplored, particularly in the context of cryptocurrency trust networks. This gap directly motivates RQ2.

3 Data Description and Preprocessing

3.1 Dataset Description

We use two publicly available signed directed graphs from the Stanford SNAP repository [5]: Bitcoin OTC and Bitcoin Alpha. Both networks model web-of-trust dynamics among anonymous users of Bitcoin trading platforms, where the absence of verifiable identity creates a concrete need for reputation mechanisms. Each directed edge $(u \rightarrow v)$ represents an explicit rating assigned by user u to user v, with integer weights in $[-10, +10]$ and an associated Unix timestamp. For the purpose of sign prediction, edge labels are binarised directly from the weight: edges with weight > 0 are labelled as trust $(+)$, and edges with weight ≤ 0 as distrust $(-)$.

Table 1: Summary statistics of the two Bitcoin web-of-trust datasets.

Dataset	Nodes	Edges	Density	Avg Deg.	Avg Clust.
Bitcoin OTC	5,881	35,592	0.001029	6.05	0.1511
Bitcoin Alpha	3,783	24,186	0.001690	6.39	0.1583

Both datasets are extremely sparse, with edge densities below 0.002, which is typical of large-scale web-of-trust networks. Despite this global sparsity, the average clustering coefficients of 0.158 (Alpha) and 0.151 (OTC) indicate that users tend to form locally dense trust communities, consistent with the prevalence of balanced triads observed in the triad census (Section 3.2.3). The near-identical average degrees (6.39 and 6.05 respectively) suggest that the underlying trust dynamics are structurally comparable across the two platforms, supporting the feasibility of cross-dataset generalisation (RQ2).

3.2 Preprocessing and Structural Analysis

Raw edges carry integer ratings in $[-10, +10]$ alongside a Unix timestamp. Two quantities are derived from each rating: a binary sign label ($+1$ if rating > 0, -1 otherwise) and an absolute weight. The resulting class distributions reveal severe imbalance in both networks: Bitcoin Alpha contains 93.6% positive and 6.4% negative edges, while Bitcoin OTC contains 90.0% positive and 10.0% negative edges.

3.2.1 Temporal Split

To preserve the causal ordering of interactions, both datasets are sorted chronologically by Unix timestamp before being partitioned into training (oldest 80%) and test (most recent 20%) sets. Although the `soc-sign-bitcoinotc` dataset was already pre-sorted, the `soc-sign-bitcoinalpha` dataset was not; programmatic sorting was therefore applied to both in a consistent manner. This partitioning yielded 19,348 / 4,838 edges for Alpha and 28,473 / 7,119 for OTC respectively.

To ensure scientific validity, we verify that no test edge precedes the last training edge, thereby preventing any form of temporal leakage. Graph objects are constructed exclusively from training edges using `NetworkX DiGraph` [6], guaranteeing that structural features computed at prediction time never incorporate future information.

3.2.2 Degree Distribution Analysis

Figure 1 reports the log-log degree distributions for both networks. In-degree and out-degree follow an approximate power-law distribution in both datasets, indicating that a small number of highly active raters and highly rated users dominate the network, a characteristic common to large-scale web-of-trust systems.

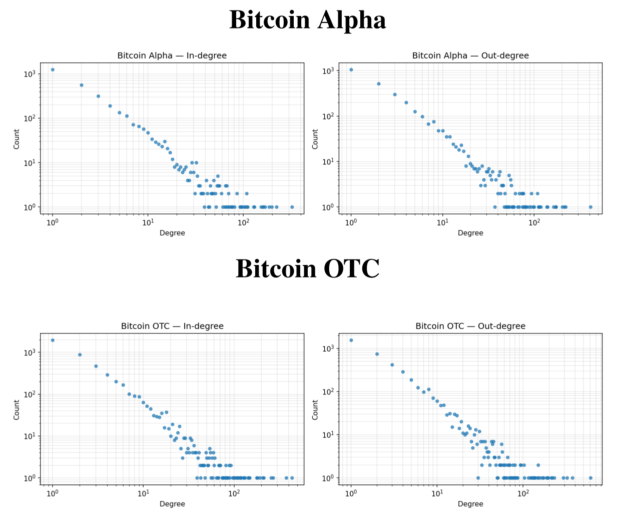

Figure 1: Log-log degree distributions (in- and out-degree) for Bitcoin Alpha and Bitcoin OTC networks.

3.2.3 Triad Census

All directed triangles in the training graphs are enumerated following the classification of Davis and Leinhardt [7] and categorised according to Balance Theory [8]. A triad is considered balanced if the product of its three edge signs is positive (i.e., $+++$ or $+--$ configurations). Bitcoin Alpha yields 49,972 triads, of which 91.0% are balanced and 9.0% unbalanced. Bitcoin OTC yields 71,020 triads, with 91.4% balanced and 8.6% unbalanced. The strong overrepresentation of balanced triads in both networks provides empirical evidence that Balance Theory actively shapes trust dynamics, and directly motivates the use of balance-theoretic features in our classifier.

4 Feature Engineering

For each candidate edge $(u \to v, t)$ in both datasets, a feature vector is constructed as input to the Random Forest classifier. The dependent variable y represents the sign of the interaction: trust $(+1)$ or distrust (-1). Every feature is computed using only edges with timestamp $< t$, so no future information leaks into the model at any stage. The feature vector is organised into three conceptual pillars: (1) local node behaviour, (2) sociological structure, and (3) reciprocity.

4.1 Design Principles

All features are computed strictly at time t using only the subgraph induced by edges that precede the candidate interaction. This constraint ensures that the classifier operates in a realistic online-prediction setting, where only historical data is available at inference time. Nodes with no prior interaction history default to zero for all features.

4.2 Local Node Behaviour Features

Local node features capture the individual reputation of each endpoint directly from their interaction history, without relying on global graph structure. For the source node u we extract three features: (1) out-degree $d^{\text{out}}(u,t)$, the total number of ratings u has issued before t; (2) mean rating given $\bar{r}^{\text{out}}(u,t)$, the average score u assigns to others; and (3) positive fraction given $\rho^{\text{out}}(u,t)$, the share of u's ratings that are positive. A negative rating from a habitually harsh rater carries less evidential weight than the same rating from a consistently positive user.

For the target node v we compute three symmetric features: (4) in-degree $d^{\text{in}}(v,t)$, the total ratings v has received; (5) mean rating received $\bar{r}^{\text{in}}(v,t)$, the average score the community assigns to v; and (6) positive fraction received $\rho^{\text{in}}(v,t)$. A node receiving 90% positive ratings is substantially more trustworthy, from the network's perspective, than one receiving only 40%.

4.3 Sociological Structure Features

4.3.1 Balance Theory Features

Balance Theory [8] predicts that triads tend toward configurations where the product of edge signs is positive. For a candidate edge $(u \to v, t)$, we identify every common neighbour w such that both $u \to w$ and $w \to v$ exist before t. For each such w, the implied sign of $u \to v$ under balance is $\hat{s} = s_{uw} \cdot s_{wv}$. We summarise this distribution with two features: the fraction of common neighbours that imply a positive sign (`frac_balance_pred_pos`) and a net balance signal `balance_signal` $= (n_+ - n_-) \,/\, n_{\text{total}} \in [-1, +1]$. When no common neighbour exists both features default to zero.

The predictive power of these features is substantiated by strong in-sample validation: among edges with at least one common neighbour, balance-only sign accuracy reaches 95.0% on the Alpha training set and 94.9% on OTC, degrading only modestly to 89.4% and 91.8% on the respective test sets. The fraction of balanced triads is 91.4% (Alpha) and 92.0% (OTC) on the training edges with at least one common neighbour, consistent with but slightly higher than the global triad census figures of 91.0% and 91.4% reported in Section 3.2.3. The small discrepancy arises because the triad census counts each triangle once, whereas the per-edge computation weights triangles by the number of candidate edges they participate in, slightly overrepresenting high-degree nodes which tend to form more balanced triads.

4.3.2 Status Theory Features

Status Theory [1] frames trust as a signal of relative social standing: assigning a positive rating implies that the target holds higher status than the source. We define the status of a node x at time t as

$$S(x,t) = d_{\text{in}}^{+}(x,t) - d_{\text{in}}^{-}(x,t), \tag{1}$$

where d_{in}^{+} and d_{in}^{-} are the positive and negative in-degrees accumulated before t. The feature for edge $(u \to v, t)$ is the status differential $\Delta S = S(v,t) - S(u,t)$: a large positive value indicates that v is substantially more reputable than u, which Status Theory associates with a positive rating.

Empirical validation supports the directional signal: on the Alpha training set, edges labelled positive carry a mean ΔS of -1.14, while edges labelled negative carry a mean of -12.08. This separation widens on the test sets (-1.26 vs. -46.86 on Alpha; -4.48 vs. -57.57 on OTC), indicating that status differentiation becomes more pronounced as the network matures.

4.4 Reciprocity Features

Bitcoin Alpha exhibits a reciprocity of 0.85 and Bitcoin OTC of 0.81, indicating that the large majority of interactions are mutual. The analysis reveals a clear asymmetry in how reciprocity behaves depending on the sign of the original edge:

- **When $u \to v$ is positive**, the reverse rating $v \to u$ is also positive in 99.0% of cases on Alpha and 98.8% on OTC — confirming that mutual trust is overwhelmingly the norm.
- **When $u \to v$ is negative**, the picture is more nuanced: the reverse rating is positive in 46.4% of cases on Alpha and 38.1% on OTC, and negative in 53.6% and 61.9% respectively. This suggests that distrust tends to be reciprocated, but far less consistently than trust.

At the edge level, reciprocity is encoded with two binary/ordinal features: `has_reverse` $\in \{0, 1\}$, which flags whether v has already rated u before t, and `reverse_sign` $\in \{-1, 0, +1\}$, which records the sign of that prior rating (0 if none exists). Approximately 40% of training edges carry a reverse counterpart in both datasets — a figure lower than the global reciprocity of $0.85/0.81$ because the latter measures whether both directions eventually exist in the full static graph, whereas `has_reverse` is computed in a streaming fashion: at the moment u rates v, the reverse rating from v may simply not have happened yet. Among edges that do carry a reverse, positive edges have a mean `reverse_sign` of $\approx$0.43, while negative edges have a mean near zero, reflecting the dominance of mutual trust over mutual distrust in the network.

4.5 Final Feature Vector

Table 2 summarises the complete set of fourteen features organised across the three pillars described above. All features are computed strictly from edges with timestamp $< t$, ensuring no future information is accessible at prediction time.

Table 2: Complete feature set used as input to the Random Forest classifier.

Pillar	Feature	Description
1. Local Node Behaviour	$d^{\text{out}}(u, t)$	Ratings u has issued before t
	$\bar{r}^{\text{out}}(u, t)$	Mean score u assigns to others
	$\rho^{\text{out}}(u, t)$	Fraction of positive ratings u gives
	$d^{\text{in}}(v, t)$	Ratings v has received before t
	$\bar{r}^{\text{in}}(v, t)$	Mean score the community assigns to v
	$\rho^{\text{in}}(v, t)$	Fraction of positive ratings v receives
2. Sociological Structure	n_{common}	Number of common neighbours w with $u \to w$ and $w \to v$
	n_+, n_-	Common neighbours implying positive / negative sign
	`frac_balance_pred_pos`	Fraction of common neighbours implying positive sign
	`balance_signal`	Net balance vote $(n_+ - n_-) / n_{\text{common}}$
	$\Delta S = S(v, t) - S(u, t)$	Status differential under Status Theory
3. Reciprocity	`has_reverse`	1 if v has rated u before t, else 0
	`reverse_sign`	Sign of v's prior rating of u; 0 if absent

5 Random Forest Algorithm

5.1 Algorithm Description

A Random Forest [9] is an ensemble learning method that constructs a collection of decision trees at training time and outputs the class predicted by the majority of those trees. Each tree is trained on a bootstrap sample of the training data (sampling with replacement), and at each internal node the best split is selected from a random subset of $\sqrt{p}$ features, where p is the total number of features, rather than from the full feature set. This double source of randomness decorrelates the individual trees and reduces variance without substantially increasing bias.

Formally, given a training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ with $y_i \in \{+1, -1\}$, the model builds T trees $\{h_t\}_{t=1}^T$ and predicts

$$\hat{y} = \operatorname{sign}\left(\sum_{t=1}^{T} h_t(\mathbf{x})\right). \tag{2}$$

The input to each tree is the feature vector described in Section 4; the output is a binary edge-sign label (trust or distrust).

Algorithm 1 Random Forest Classifier for Edge-Sign Prediction

Require: Training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, number of trees T, feature vector $\mathbf{x}_i \in R^p$
Ensure: Predicted edge-sign label $\hat{y} \in \{+1, -1\}$
for $t = 1$ **to** T **do**
  Draw a bootstrap sample $\mathcal{D}_t$ from $\mathcal{D}$ with replacement
  At each node, select $\lfloor \sqrt{p} \rfloor$ features at random
  Grow tree h_t by recursively splitting on the most informative feature, until a stopping criterion is met
end for
return $\hat{y} = \operatorname{sign}\left(\sum_{t=1}^{T} h_t(\mathbf{x})\right)$

5.2 Training Setup and Hyperparameter Search

To select the optimal configuration we performed a grid search over the hyperparameter space shown in Table 3 using `GridSearchCV` with 5-fold `StratifiedKFold` cross-validation, optimising for AUC-ROC. The best configuration selected for each dataset is reported in Table 4.

Table 3: Hyperparameter grid explored during cross-validation.

Hyperparameter	Values explored
`n_estimators`	300, 500
`max_depth`	None (unlimited), 10, 20
`min_samples_leaf`	1, 5
`max_features`	`sqrt`

Dataset	n_est.	max_depth	min_leaf	**CV AUC**
Bitcoin Alpha	500	10	5	0.9220
Bitcoin OTC	500	10	5	0.9481

Table 4: Best hyperparameters selected by 5-fold CV (AUC-ROC scoring).

5.3 Handling Class Imbalance

The `class_weight='balanced'` option is applied in all configurations to counteract the severe class imbalance documented in Section 3: each tree reweights the loss so that misclassifying a distrust edge is penalised proportionally to its underrepresentation (93.6% positive edges on Alpha, 90.0% on OTC). All experiments use `random_state=42` for reproducibility.

6 Signed Graph Convolutional Network

6.1 Architecture

The SGCN proposed by Derr et al. [3] extends spectral graph convolution to signed graphs by maintaining two separate aggregation channels at each layer: one for nodes reachable along *balanced paths* (suggested friends under Balance Theory) and one for nodes reachable along *unbalanced paths* (suggested enemies). Formally, for node u_i the balanced representation at layer l is

$$\mathbf{h}_i^{B(l)} = \sigma\left(\mathbf{W}^{B(l)}\left[\frac{\sum_{j\in\mathcal{N}_i^+}\mathbf{h}_j^{B(l-1)}}{|\mathcal{N}_i^+|},\ \frac{\sum_{k\in\mathcal{N}_i^-}\mathbf{h}_k^{U(l-1)}}{|\mathcal{N}_i^-|},\ \mathbf{h}_i^{B(l-1)}\right]\right) \tag{3}$$

and symmetrically for the unbalanced representation $\mathbf{h}_i^{U(l)}$, where $\mathcal{N}_i^+$ and $\mathcal{N}_i^-$ denote the positive and negative neighbour sets of u_i, and $\mathbf{W}^{B(l)}$, $\mathbf{W}^{U(l)}$ are learnable weight matrices. The final node embedding concatenates both representations: $\mathbf{z}_i = [\mathbf{h}_i^{B(L)}, \mathbf{h}_i^{U(L)}]$. We use the PyTorch Geometric implementation [4] with $L = 2$ convolutional layers, hidden dimension 64, and balance regularisation parameter $\lambda = 5$.

6.2 Training Setup

Following Derr et al., the graph is treated as undirected: each directed training edge $u \to v$ is inserted in both directions into the appropriate positive or negative adjacency matrix. The model is trained on the same temporal training split used for the Random Forest, so graph structure never incorporates future edges. Node input features are the L2-normalised six-dimensional local behaviour statistics described in Section 4.2, computed from training history only; nodes unseen during training are initialised to zero. The model is optimised with Adam (lr $= 0.01$, weight decay 5×10^{-4}) for 300 epochs. Sign prediction for a test edge (u, v) is determined by the sign of the dot product $\mathbf{z}_u \cdot \mathbf{z}_v$ between the learned embeddings.

6.3 Evaluation and Comparison with Random Forest

Table 5 compares the two models on the held-out test sets using the same metrics and temporal splits.

Table 5: Random Forest vs. SGCN on the held-out test sets. $F1^-$: distrust class; $F1^+$: trust class.

Model	**Dataset**	**AUC-ROC**	$F1^-$	$F1^+$
Random Forest	Bitcoin Alpha	0.8665	0.4924	0.9064
SGCN	Bitcoin Alpha	0.5478	0.0000	0.9319
Random Forest	Bitcoin OTC	0.9106	0.6412	0.9191
SGCN	Bitcoin OTC	0.6578	0.0000	0.9167

The SGCN achieves AUC-ROC of 0.5478 on Alpha and 0.6578 on OTC, substantially below the Random Forest on both datasets. On both networks the model collapses to predicting trust for every edge ($F1^- = 0$), meaning no distrust edge is ever detected. The positive AUC scores above 0.5 indicate that the learned embeddings retain some weak discriminative ability (positive edges tend to receive slightly higher dot-product scores than negative ones) but the scores never cross zero, so no negative prediction is ever issued at inference time.

This collapse is a direct consequence of the severe class imbalance during training. The positive aggregation channel is exposed to roughly 18,000 and 26,005 training edges respectively, while the negative channel sees only 919 and 2,468. With an order of magnitude fewer examples, the negative channel cannot learn embeddings that consistently point away from the positive cluster, so all dot products remain positive regardless of the true edge sign.

We note that Derr et al. report AUC of 0.796 and 0.823 on the same datasets; that evaluation uses a random 80/20 split rather than a temporal one. Under random split, test edges are distributed throughout the network's history and share neighbourhood structure with training edges, making the task substantially easier. Our temporal split requires the model to generalise to an evolved network where new users and interaction patterns have emerged after the training cutoff, which constitutes a more realistic but harder evaluation protocol.

7 Results

This section presents the experimental evaluation of the Random Forest classifier across four conditions. Sections 7.1–7.3 jointly address RQ1 by examining in-dataset performance, feature importances, and the marginal contribution of each theoretical pillar. Section 7.4 completes RQ1 by comparing the Random Forest against the SGCN. Section 7.5 addresses RQ2 through cross-dataset transfer evaluation.

7.1 In-Dataset Classification Performance on Both Platforms

Each model is trained on the temporally ordered training split (oldest 80%) and evaluated on the held-out test set (most recent 20%), following the partitioning described in Section 3 and the training protocol of Section 5.2. Table 6 reports AUC-ROC and F1-score on both classes for Bitcoin Alpha and Bitcoin OTC.

Table 6: Random Forest test-set performance on both datasets. $F1^-$ denotes F1-score on the distrust (minority) class; $F1^+$ on the trust class.

Dataset	CV AUC	Test AUC	$F1^-$	$F1^+$
Bitcoin Alpha	0.9220	0.8665	0.4924	0.9064
Bitcoin OTC	0.9481	0.9106	0.6412	0.9191

AUC-ROC is the primary metric because it is threshold-independent and robust to class imbalance. $F1^-$ on the distrust class ($y = -1$) is the secondary metric of interest, as it directly measures the model's ability to detect potentially fraudulent interactions. The gap between CV AUC-ROC and test AUC-ROC reflects the increasing difficulty of predicting edges that occur later in the network's lifetime, when new users appear and trust dynamics evolve beyond the patterns observed during training.

It is worth noting that $F1^-$ of 0.49 on Alpha does not imply the model ignores distrust entirely. The underlying confusion matrix shows a recall of 60.1% on the distrust class (371 out of 617 negative test edges correctly identified), at the cost of 519 false positives. The low $F1^-$ is therefore driven primarily by limited precision (41.7%) rather than by an inability to detect fraud, and reflects the structural difficulty of distinguishing a rare minority class whose members share many surface-level features with the majority.

7.2 Feature Importances

Figure 2 reports the mean impurity decrease (Gini importance) for each feature, averaged across all trees, for both datasets. Colours reflect the feature pillar each variable belongs to: local node behaviour (blue), sociological structure (orange), and reciprocity (green).

Gini importances should be interpreted with two caveats. First, a near-binary feature such as `reverse_sign` is engaged at relatively few split nodes across all trees, which mechanically lowers its Gini score even when it is highly informative at the nodes where it is used. This explains why reciprocity appears modest in Figure 2 despite causing the largest AUC drop when removed in the ablation study (Section 7.3). Second, when two features carry overlapping information, the trees distribute their usage across both, so each receives only partial credit; `status_diff` is largely redundant with local node behaviour features and, as the ablation confirms, its removal has negligible impact. For these reasons the Gini plot is most useful for ranking individual variables, while the ablation study provides the primary evidence for comparing pillar-level contributions.

Local node behaviour features dominate the ranking. `v_pos_frac` and `v_mean_received` rank among the top predictors in both datasets, reflecting that a node's historical reception by the community is the strongest individual signal of its future trustworthiness. Among the sociological pillars, `frac_balance_pred_pos` and `balance_signal` carry substantial importance, consistent with the 91–92% balanced triad rate observed in both training graphs.

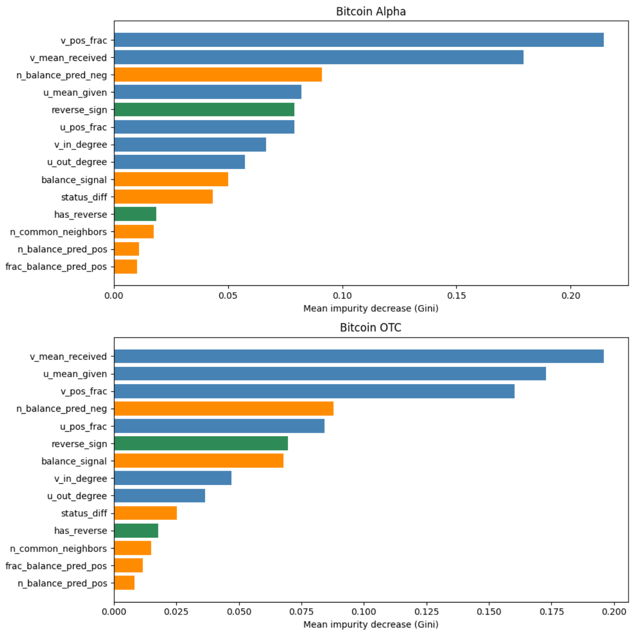

Figure 2: Random Forest feature importances for Bitcoin Alpha (top) and Bitcoin OTC (bottom), sorted in descending order.

`status_diff` ranks below the Balance Theory features, suggesting that relative social standing is a weaker predictor than local triad structure in these networks.

7.3 Ablation Study

To quantify the marginal contribution of each feature pillar, the Random Forest classifier is systematically retrained under four distinct configurations. In each run, one specific feature group is excluded while all other variables are retained, keeping the hyperparameters strictly fixed to the optimal values selected for the full model (Table 4). This controlled ablation approach ensures that any observed drop in classification performance (AUC-ROC) is isolated and attributable solely to the removed feature domain.

The analysis evaluates the following four conditions: (1) with local node behaviour features only; (2) without Balance Theory features; (3) without Status Theory features; and (4) without reciprocity features. The full model, trained simultaneously on all feature pillars, serves as the reference baseline throughout the analysis.

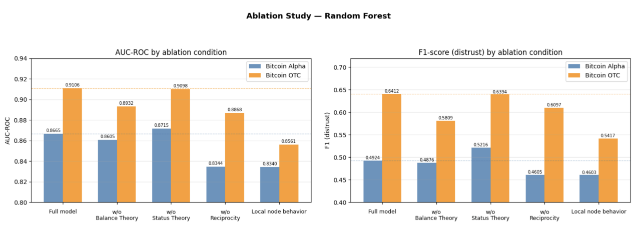

Figure 3: AUC-ROC (left) and F1-score on the distrust class (right) for each ablation condition on Bitcoin Alpha and Bitcoin OTC. Dashed lines mark the full-model performance for each dataset.

Several consistent patterns emerge across both datasets. Reciprocity is the most impactful pillar overall: its removal

Table 7: Ablation study results. Each condition removes one feature pillar while keeping all others. $F1^-$: distrust class; $F1^+$: trust class.

Condition	AUC-ROC	$F1^-$	$F1^+$
Bitcoin Alpha			
Full model	0.8665	0.4924	0.9064
w/o Balance Theory	0.8605	0.4876	0.8979
w/o Status Theory	0.8715	0.5216	0.9072
w/o Reciprocity	0.8344	0.4605	0.9075
Local node behaviour only	0.8340	0.4603	0.8921
Bitcoin OTC			
Full model	0.9106	0.6412	0.9191
w/o Balance Theory	0.8932	0.5809	0.8895
w/o Status Theory	0.9098	0.6394	0.9178
w/o Reciprocity	0.8868	0.6097	0.9129
Local node behaviour only	0.8561	0.5417	0.8758

causes the largest AUC drop across all conditions, -0.0321 on Alpha and -0.0238 on OTC, consistent with the near-perfect concordance rates between reciprocal edges reported in Section 4.4. Balance Theory is the most informative sociological pillar; removing it produces the largest single-pillar AUC drop in both datasets (-0.0060 on Alpha, -0.0174 on OTC), with an even more pronounced effect on $F1^-$, which falls by 0.0048 and 0.0603 respectively. Status Theory contributes marginally: on Alpha its removal yields a slight AUC increase ($+0.0050$), and on OTC the drop is only -0.0008, confirming that `status_diff` encodes little independent information beyond the local node behaviour features. This is consistent with the construction of the feature: status is defined as the difference between positive and negative in-degree, quantities that the local node behaviour pillar already captures through $d^{\text{in}}(v,t)$ and $\rho^{\text{in}}(v,t)$. Once those features are available, the forest can approximate the status differential without observing it directly, making the explicit Status Theory feature largely superfluous. Finally, retaining only local node behaviour reduces AUC to 0.8340 on Alpha and 0.8561 on OTC, drops of -0.0325 and -0.0545 respectively, confirming the value of the full feature set.

7.4 Random Forest versus Signed Graph Convolutional Network

The comparison between the Random Forest and the SGCN is reported in Table 5 in Section 6. The Random Forest outperforms the SGCN by a large margin on both AUC-ROC ($+0.3187$ on Alpha, $+0.2528$ on OTC) and $F1^-$, with the SGCN issuing zero distrust predictions on both datasets. The hand-crafted feature pipeline therefore provides a more effective inductive bias for sign prediction on these sparse, heavily imbalanced networks than learned graph representations. The explicit encoding of Balance Theory, reciprocity, and node behaviour captures structural invariants that the SGCN's message-passing mechanism cannot recover from the limited local neighbourhood information available in graphs with density below 0.002.

7.5 Cross-Dataset Transfer

To address RQ2, the two trained Random Forest models are applied directly to the test set of the opposing dataset without any retraining or hyperparameter retuning, yielding two transfer directions: OTC $\rightarrow$ Alpha and Alpha $\rightarrow$ OTC. Table 8 reports results alongside the in-dataset baselines for direct comparison.

Direction	**AUC-ROC**	**F1$^-$**	**F1$^+$**
Alpha $\rightarrow$ Alpha (in-dataset)	0.8665	0.4924	0.9064
OTC $\rightarrow$ Alpha (cross-dataset)	0.8835	0.5608	0.9309
OTC $\rightarrow$ OTC (in-dataset)	0.9106	0.6412	0.9191
Alpha $\rightarrow$ OTC (cross-dataset)	0.8779	0.5404	0.8743

Table 8: Cross-dataset generalisation results.

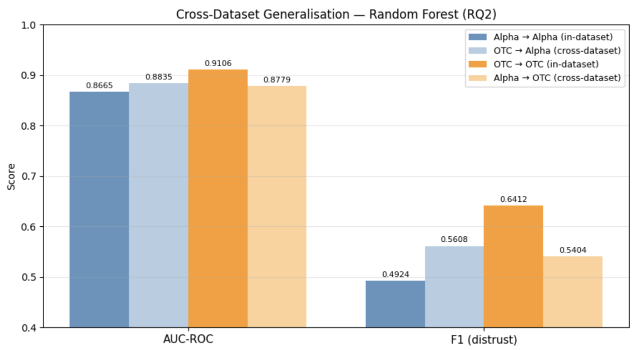

Figure 4: AUC-ROC and F1-score (distrust class) for in-dataset and cross-dataset evaluation. Full bars indicate in-dataset performance; faded bars indicate cross-dataset transfer.

Both transfer directions maintain AUC-ROC above 0.87, well above random performance, providing a positive answer to RQ2. The OTC $\rightarrow$ Alpha direction yields a cross-dataset AUC of 0.8835, which not only preserves but exceeds the in-dataset Alpha baseline of 0.8665. This can be attributed to the larger OTC training set (28,473 edges vs. 19,348 for Alpha), which provides a richer signal for learning trust dynamics, and to OTC's higher distrust rate (10.0% vs. 6.4%), which better exposes the model to negative edge patterns. The Alpha $\rightarrow$ OTC direction shows a modest degradation in AUC (0.8779 vs. 0.9106 in-dataset) and $F1^-$ (0.5404 vs. 0.6412), as expected given that Alpha is the smaller dataset with fewer negative examples. The gap nevertheless remains contained, consistent with the near-identical density and average degree of the two networks reported in Table 1.

8 Discussion

8.1 Sociological Features, Reciprocity, and the Predictive Power of Structural Patterns (RQ1)

The ablation study provides a clear answer to the first part of RQ1. Balance Theory and reciprocity offer consistent and meaningful predictive lift over local node behaviour alone. Reciprocity is the single most impactful pillar, driven by the near-perfect sign concordance between reciprocal edges observed in both networks. Balance Theory is the strongest sociological signal, particularly for distrust detection, as evidenced by the pronounced drops in $F1^-$ when it is removed. Status Theory, by contrast, proves largely redundant with local node behaviour: `status_diff` captures little independent information once degree and mean-rating features are already present, and its removal has negligible effect on performance.

The comparison with the SGCN completes the answer to RQ1. Despite encoding balance-theoretic assumptions directly into its architecture, the SGCN achieves AUC of only 0.55 and 0.66 on Alpha and OTC respectively, issuing zero distrust predictions on both datasets. This result demonstrates that on sparse, heavily imbalanced graphs, the explicit feature engineering approach is more effective than learned representations: the hand-crafted features distil decades of sociological theory into a compact and robust signal, whereas the SGCN's message-passing mechanism is starved of the local neighbourhood density it requires to propagate meaningful information.

8.2 Structural Comparability and Cross-Platform Generalisation (RQ2)

The cross-dataset evaluation provides a positive answer to RQ2: trust dynamics learned on one Bitcoin platform generalise to the other without any retraining. Both transfer directions maintain AUC-ROC above 0.87, and the OTC $\rightarrow$ Alpha direction even surpasses the Alpha in-dataset baseline. This generalisation is not coincidental. As shown in Table 1, the two networks share near-identical structural properties: average degrees of 6.05 and 6.39, clustering coefficients of 0.151 and 0.158, and comparable edge densities. These similarities suggest that the same underlying trust formation process governs both platforms, and that the hand-crafted features encode structural invariants that are not specific to either network.

8.3 Limitations

Three limitations merit acknowledgement. First, $F1^-$ on Bitcoin Alpha remains below 0.50 even for the full model, reflecting a ceiling imposed by severe class imbalance and the structural overlap between minority and majority classes that feature-based classifiers cannot fully resolve. Second, Gini importance rankings should not be used as the sole criterion for evaluating feature contributions, as discussed in Section 7.2; the ablation study provides more reliable pillar-level evidence. Third, the SGCN evaluation is conducted under a temporal split that differs from the random split used by Derr et al., making direct numerical comparison with published results inadvisable. While this protocol is methodologically more rigorous, it prevents a clean benchmark against the original paper and leaves open the question of how much of the observed performance gap is attributable to model capacity versus evaluation difficulty.

8.4 Future Work

Several directions remain open. The SGCN evaluation protocol could be extended by adopting negative-edge oversampling or signed attention mechanisms to mitigate the class imbalance in the negative aggregation channel. Temporal graph neural networks, which explicitly model edge timestamps rather than treating the graph as static, represent a promising alternative to the current approach. On the feature side, incorporating higher-order structural motifs beyond triangles, or community-level features derived from signed spectral clustering, could further improve distrust detection. Finally, the cross-dataset transfer experiment could be extended to other signed networks such as Epinions or Slashdot to test whether the structural generalisability observed between Bitcoin OTC and Alpha holds more broadly.

9 Conclusion

We have investigated sign prediction in Bitcoin web-of-trust networks as a binary edge-sign classification problem under severe class imbalance. A feature-engineering pipeline grounded in Balance Theory, Status Theory, and reciprocity, computed via a temporally strict streaming procedure, yields a Random Forest classifier that achieves AUC-ROC of 0.8665 on Bitcoin Alpha and 0.9106 on Bitcoin OTC. The ablation study directly answers RQ1: reciprocity is the single most impactful feature pillar, followed by Balance Theory, while Status Theory contributes marginally and redundantly with local node behaviour features. A SGCN trained on the same splits achieves an AUC of only 0.55–0.66 with $F1^- = 0$ on both datasets, confirming that on sparse, heavily imbalanced signed graphs, explicit sociological feature engineering outperforms learned graph representations under realistic temporal evaluation conditions. Cross-dataset transfer maintains AUC-ROC above 0.87 in both directions without retraining, providing a positive answer to RQ2 and confirming that the structural similarity between the two platforms (near-identical density, average degree, and clustering coefficient) reflects a shared underlying trust formation process that the feature-based model successfully captures. This transferability is, however, contingent on structural comparability: whether it extends to signed networks with substantially different topology or class distributions remains an open question for future work.

Code Availability

The source code developed for this study is publicly available in the project repository [10].

Use of Generative AI

We used Claude (Anthropic) to support understanding of course concepts, to verify the correctness of implementation logic and check for potential sources of data leakage, and to assist in refining the language of the text for clarity and academic register. All experimental design decisions, mathematical formulations, implementations, and interpretations of results are our own.

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